

Binary & Binary Addition (Lecture Notes)

1. Introduction

- Computers use **binary (0s and 1s)** → electricity (on/off).
 - Analogy: light switch, true/false.
 - **Question:** What else in life has only two states?
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2. Number Systems

Decimal (Base 10)

- Digits: 0–9
- Place values: 1, 10, 100...

Binary (Base 2)

- Digits: 0–1
 - Place values: 1, 2, 4, 8...
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3. Conversion Example (13 → binary)

$$13 \div 2 = 6 \text{ remainder } 1$$

$$6 \div 2 = 3 \text{ remainder } 0$$

$$3 \div 2 = 1 \text{ remainder } 1$$

$$1 \div 2 = 0 \text{ remainder } 1$$

Answer: 1101

4. Quick Activity

Convert:

- $5 \rightarrow ?$
 - $10 \rightarrow ?$
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5. Binary Addition Rules

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 10 \text{ (carry the 1)}$$

6. Examples

Example 1

$$\begin{array}{r} 101 \\ + 011 \\ \hline 1000 \end{array}$$

Example 2

$$\begin{array}{r} 111 \\ + 1 \\ \hline 1000 \end{array}$$

7. Practice Problem

Solve:

```
  1011
+ 1101
-----
  ????
```

8. Overflow

What happens when result doesn't fit?

```
  1111
+   1
-----
10000
```

9. Bigger Picture

- Binary addition = foundation of computing.
- Logic gates implement addition:
 - XOR → sum
 - AND → carry
- Adders scale up → CPUs.

10. Wrap-Up

- Binary = computer's language.
- Addition = simple rules with carry.
- Foundation for everything (text, images, processing).

Class Question: Where else do you think binary shows up in computing?